

Resume Keyword Analyzer

1. Introduction

Resume Keyword Analyzer is a simple Python project. The main purpose of this project is to compare a person's resume with a job description.

When a person applies for a job, the company gives some requirements and skills in the job description. The candidate also has different skills mentioned in the resume. This project checks both texts and finds which skills are matched and which skills are missing.

The system also calculates a keyword coverage percentage. This percentage shows how much the resume matches the job requirements.

For example, if a job requires **Python, SQL, Machine Learning, TensorFlow, and Docker**, and the resume contains **Python, SQL, and Machine Learning**, then the system will show the matched and missing skills.

2. Problem Understanding

Many job descriptions contain different required skills. A recruiter needs to check whether the candidate has those skills or not.

Checking every resume manually can take time. This project provides a simple solution for comparing a resume with a job description.

The project answers the following questions:

- Which skills are present in both the resume and job description?
- Which required skills are missing from the resume?
- How much does the resume match the job description?

The system gives the result in a simple and clear format.

3. Project Objectives

The main objectives of this project are:

- To take resume text as input.
- To take job description text as input.
- To find important skills from the text.

- To compare resume skills with job requirements.
 - To show matched skills.
 - To show missing skills.
 - To calculate the keyword coverage percentage.
 - To generate a simple analysis report.
-

4. Proposed Approach

The project works in different steps.

First, the user provides the resume text and the job description text.

After that, the text is cleaned and converted into lowercase.

Then, the system checks the skills from a predefined list.

The skills found in the resume and job description are compared.

Finally, the system shows:

- Matched skills
- Missing skills
- Keyword coverage percentage

5. Implementation

The project is developed using **Python**.

The project is created and tested in **Google Colab**.

Google Colab is used because it allows us to write and run Python code online.

Tools Used

- Python
- Google Colab
- Regular Expressions (Regex)
- Python Lists
- Python Sets

The project uses a list of common technical skills. Some examples are:

- Python
- SQL
- Machine Learning
- Deep Learning

- Pandas
- NumPy
- Scikit-learn
- TensorFlow
- Keras
- PyTorch
- Computer Vision
- OpenCV
- Flask
- Docker
- AWS
- Git
- GitHub

The system checks these skills in the resume and job description.

6. Text Preprocessing

Before comparing the text, some simple preprocessing is performed.

Convert Text to Lowercase

The text is converted to lowercase.

For example:

```
Python ->python  
PYTHON -> python  
Python -> python
```

This helps the system treat different capital letters as the same word.

Remove Extra Spaces

Extra spaces are removed from the text.

This makes the text cleaner and easier to compare.

7. Skill Extraction

The project has a predefined list of technical skills.

The system checks whether these skills are present in the resume and the job description.

For example, the job description may contain:

We need a candidate with Python, SQL, Machine Learning, and TensorFlow skills.

The system finds:

- Python
- SQL
- Machine Learning
- TensorFlow

The same process is used for the resume.

8. Skill Comparison

After finding the skills, the system compares the resume skills with the job skills.

The result is divided into two parts.

Matched Skills

These are the skills found in both the resume and the job description.

For example:

- Python
- SQL
- Machine Learning

Missing Skills

These are the skills required in the job description but not found in the resume.

For example:

- TensorFlow
- Docker

9. Keyword Coverage Calculation

The system also calculates the keyword coverage percentage.

The formula is:

Keyword Coverage = (Matched Skills / Total Required Skills) × 100

10. Important Technical Decisions

Predefined Skills List

A predefined list of technical skills is used in this project.

The system checks these skills in both the resume and the job description.

Lowercase Matching

All text is converted into lowercase before comparison.

This helps avoid problems with capital and small letters.

Python Sets

Python sets are used to compare the skills.

Sets make it easy to find common skills and missing skills.

Clear Results

The project does not only show a percentage.

It also shows the matched and missing skills. This helps the user understand the result easily.

11. Testing Performed

Different test cases were used to test the project.

Test Case 1

Resume:

Python, SQL, Machine Learning, Pandas

Job Description:

Python, SQL, Machine Learning, TensorFlow, Docker

Expected Result:

Matched Skills:

- Python
- SQL

- Machine Learning

Missing Skills:

- TensorFlow
- Docker

Keyword Coverage:

60%

Test Case 2**Resume:**

Python, SQL, Machine Learning, Pandas, Scikit-learn

Job Description:

Python, SQL, Machine Learning, Pandas, Scikit-learn

Expected Result:

All skills are matched.

Missing Skills:

None

Keyword Coverage:

100%

Test Case 3**Resume:**

HTML, CSS, JavaScript

Job Description:

Python, SQL, Machine Learning

Expected Result:**Matched Skills:**

None

Missing Skills:

- Python
- SQL
- Machine Learning

Keyword Coverage:

0%

12. Challenges Encountered

Some problems were faced while making the project.

Different Names of the Same Skill

Some skills can be written in different ways.

For example:

- Scikit-learn
- Scikit Learn
- sklearn

The system may consider them different.

Short Forms

Some skills have short names.

For example:

- NLP = Natural Language Processing
- ML = Machine Learning

The basic system may not understand that these are the same.

Multi-Word Skills

Some skills contain more than one word.

For example:

- Machine Learning
- Deep Learning
- Data Analysis
- Computer Vision

These skills need to be checked as complete phrases.

Limited Skills List

The system only checks the skills available in the predefined list.

If a skill is not in the list, the system may not find it.

13. Solutions Implemented

Text Cleaning

The text is converted into lowercase and extra spaces are removed.

This makes the comparison better.

Complete Phrase Matching

The system checks complete skills such as:

- Machine Learning
- Deep Learning
- Computer Vision

This helps in finding multi-word skills.

Clear Report

The project shows the final result in a simple format.

The user can easily see which skills are matched and which skills are missing.

14. Results

The Resume Keyword Analyzer successfully compares the resume with the job description.

The system provides three main results:

1. Matched Skills
2. Missing Skills
3. Keyword Coverage Percentage

Example Result

RESUME KEYWORD ANALYSIS REPORT

Total Required Skills: 5

Matched Skills: 3

Missing Skills: 2

Keyword Coverage: **60%**

Matched Skills

- Python
- SQL
- Machine Learning

Missing Skills

- TensorFlow
- Docker

The result helps the user understand how well the resume matches the job description.

15. Future Scope

The project can be improved in the future.

Some possible improvements are:

PDF Upload

Users can upload their resume in PDF format.

DOCX Support

The system can also support Microsoft Word resume files.

Automatic Skill Extraction

The system can automatically find skills from the job description.

Better NLP

Advanced NLP techniques can be used to understand similar words and skills.

For example:

- NLP and Natural Language Processing
- ML and Machine Learning

Suggestions

The system can suggest skills that are missing from the resume.

Charts

Charts can be added to show:

- Matched skills
- Missing skills
- Keyword coverage

Web Application

The project can later be converted into a web application using tools such as Streamlit or Flask.

16. Conclusion

The Resume Keyword Analyzer is a simple project made using Python.

The project compares a resume with a job description and finds the skills that are matched and missing.

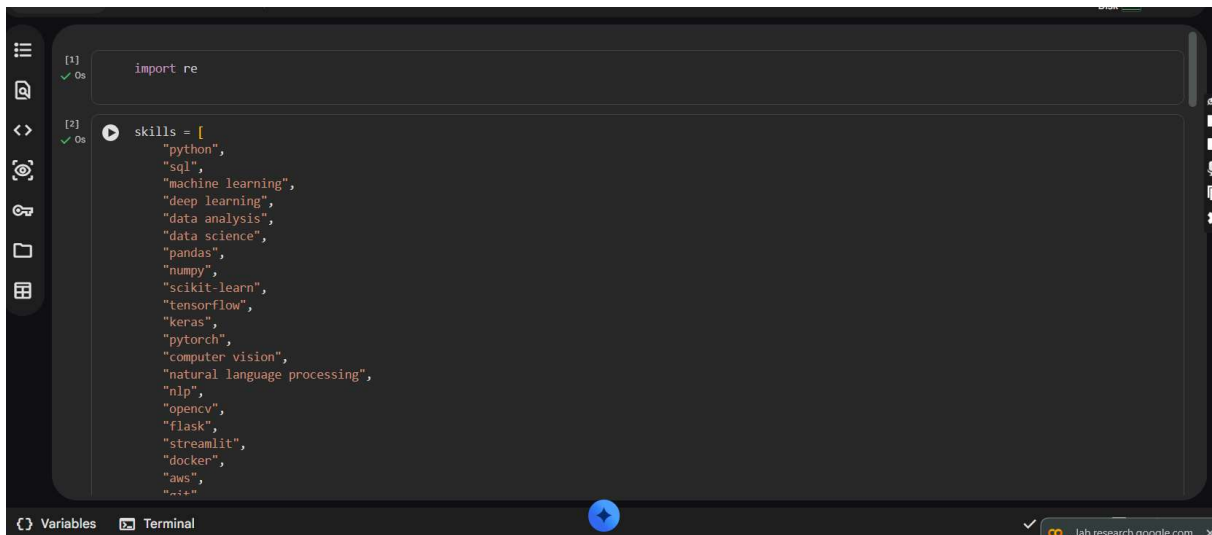
It also calculates a keyword coverage percentage.

This project shows the use of Python and basic text processing in a real-world problem.

The current project uses simple keyword matching, but it can be improved in the future by adding advanced NLP, file upload, automatic skill extraction, and better suggestions.

Overall, the project provides a simple way to understand how well a resume matches a job description.

17. ScreenShots Of Code and Result



```
[1] ✓ 0s
import re

[2] ✓ 0s
skills = [
    "python",
    "sql",
    "machine learning",
    "deep learning",
    "data analysis",
    "data science",
    "pandas",
    "numpy",
    "scikit-learn",
    "tensorflow",
    "keras",
    "pytorch",
    "computer vision",
    "natural language processing",
    "nlp",
    "opencv",
    "flask",
    "streamlit",
    "docker",
    "aws",
    "aws"
]
```

```
resume = """
I am a Software Engineering graduate with knowledge of Python,
SQL, Machine Learning, Pandas, NumPy, Scikit-learn and
Computer Vision. I have worked on Machine Learning and
Deep Learning projects.
"""
```

```
job_description = """
We are looking for a Machine Learning Intern with knowledge of
Python, SQL, Machine Learning, Pandas, Scikit-learn,
TensorFlow, Docker and AWS.
"""
```

```
def preprocess_text(text):

    # Convert text to lowercase
    text = text.lower()

    # Remove extra spaces
    text = re.sub(r'\s+', ' ', text)

    return text
```

```
def extract_skills(text, skills_list):

    found_skills = []

    for skill in skills_list:

        if skill in text:
            found_skills.append(skill)

    return found_skills
```

```
clean_resume = preprocess_text(resume)
clean_job_description = preprocess_text(job_description)

resume_skills = extract_skills(clean_resume, skills)

job_skills = extract_skills(clean_job_description, skills)

matched_skills = list(set(resume_skills) & set(job_skills))

missing_skills = list(set(job_skills) - set(resume_skills))
```

```
if len(job_skills) > 0:
    coverage = (len(matched_skills) / len(job_skills)) * 100
else:
    coverage = 0
```

```
print("=" * 50)
print("RESUME KEYWORD ANALYSIS REPORT")
print("=" * 50)

print("\nTotal Required Skills:", len(job_skills))
print("Matched Skills:", len(matched_skills))
print("Missing Skills:", len(missing_skills))

print(f"\nKeyword Coverage: {coverage:.2f}%")

print("\nMATCHED SKILLS")

if matched_skills:
    for skill in matched_skills:
        print("✓", skill.title())
else:
    print("No matched skills found.")

print("\nMISSING SKILLS")

if missing_skills:
    for skill in missing_skills:
        print("X", skill.title())
else:
    print("No missing skills found.")
```

RESUME KEYWORD ANALYSIS REPORT

Total Required Skills: 8
Matched Skills: 5
Missing Skills: 3

Keyword Coverage: 62.50%

MATCHED SKILLS

✓ Pandas
✓ Machine Learning
✓ Scikit-Learn
✓ Sql
✓ Python

Result

MISSING SKILLS

X Docker
X Tensorflow
X Aws